Session 5 | Probability & Statistics for Modelling

Solution sheet

Worked solutions with short reminders of the key idea. Numerical answers are boxed; the conceptual parts are meant to be talked through together at the end of the session.

Exercise 1 *Name the distribution, mean and variance.*

Solution.

(a) Binomial$(12, 0.5)$: mean $np = \boxed{6}$, variance $np(1-p) = \boxed{3}$.

(b) Poisson(2): mean $=$ variance $= \boxed{2}$.

(c) $\boxed{\text{Continuous}}$. A Normal is natural because height is a sum of many small, roughly independent influences, and the central limit theorem drives such sums toward a bell shape.

(d) The $\boxed{\text{Poisson}}$: for it mean $=$ variance $= \lambda$. The Binomial has variance $np(1-p) < np =$ mean, so "variance equals mean" is the Poisson's fingerprint.

Reminder

Binomial counts successes in a fixed number of trials; Poisson counts rare events in a window (and mean $=$ variance); the Normal models continuous quantities built from many small effects.

Exercise 2 *Density versus mass.*

Solution.

(a) $\mathbb{P}(X = 5) = \boxed{0}$: a single point carries no probability for a continuous variable.

(b) $\boxed{\text{Yes}}$. A density is not a probability; only its *area* must equal 1, so $f(x)$ may exceed 1 over a short interval.

(c) The value 0.5 is a density *height*, not the probability of a point. The correct 50% statement is about an *interval*: any sub-interval of length 1 carries mass 0.5, e.g. $\mathbb{P}(0 \le X \le 1) = 0.5$.

Reminder

Probability is *area under the density*: $\mathbb{P}(a \le X \le b) = \int_a^b f(x)\,dx$, and $\mathbb{P}(X = x) = 0$ for any single point.

Exercise 3 $X \in \{0, 1, 2\}$ *with probabilities* $0.2, 0.5, 0.3$.

Solution.

(a) $\mathbb{E}[X] = 0(0.2) + 1(0.5) + 2(0.3) = \boxed{1.1}$.

(b) $\mathbb{E}[X^2] = 0(0.2) + 1(0.5) + 4(0.3) = 1.7$, so $\mathrm{Var}(X) = 1.7 - 1.1^2 = \boxed{0.49}$.

(c) $\mathrm{SD}(X) = \sqrt{0.49} = \boxed{0.7}$: the typical distance of a value from the mean 1.1.

(d) A larger variance means a more spread-out distribution, so its standard deviation is $\boxed{\text{larger}}$, even though the balance point (mean) is the same.

Reminder

$\mathbb{E}[X] = \sum_x x\,p(x)$ is the balance point; $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ is the average squared spread; SD puts that spread back in the original units.

Exercise 4 $\mathrm{Var}(X) = 4$, $\mathbb{E}[X] = 5$.

Solution.

(a) $\mathrm{Var}(2X - 1) = 2^2\,\mathrm{Var}(X) = \boxed{16}$; $\mathbb{E}[3X + 2] = 3(5) + 2 = \boxed{17}$.

(b) Independent, so $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) = \boxed{13}$.

(c) A shift moves every value equally, so distances from the mean (and hence the variance) are unchanged; a scale multiplies those distances, and variance squares the factor (a^2). Variances add *only* when there is no covariance, which independence guarantees.

Reminder

$\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$; variances of independent variables add; expectation is linear always.

Exercise 5 *Scores* $\sim \mathcal{N}(100, 15^2)$.

Solution.

(a) $z = \dfrac{115 - 100}{15} = \boxed{1}$.

(b) 130 is $z = 2$, so $\mathbb{P}(X > 130) \approx \boxed{2.3\%}$.

(c) Middle 95% lies within $\pm 2\sigma$: $\boxed{[70,\ 130]}$.

(d) *Different* raw scores (one standard deviation is a different raw amount on each test), but the *same* percentile ($\approx$ 84th), because the z-score, and therefore the area below it, is identical.

Reminder

The z-score $\frac{X-\mu}{\sigma}$ expresses a value in standard-deviation units, so it is comparable across scales; recall 68–95–99.7 for $\pm 1, 2, 3\sigma$.

Exercise 6 *Population* $\sigma = 12$.

Solution.

(a) $\text{SE} = \dfrac{12}{\sqrt{36}} = \boxed{2}$.

(b) $\dfrac{12}{\sqrt{n}} = 1 \Rightarrow n = \boxed{144}$.

(c) Only the $\boxed{\text{standard error}}$ shrinks; σ is a fixed property of the population and does not change with n. More data sharpens the *mean*, not the spread of individuals.

(d) The central limit theorem is about the *average*, not single values. Averaging cancels the ups and downs, so the mean becomes Normal even when one measurement is far from Normal.

Reminder

$\text{SE} = \sigma/\sqrt{n}$: precision improves like $\sqrt{n}$, so four times the data halves the error, while σ stays put.

Exercise 7 *Data* $\{2, 4, 6, 8, 10\}$.

Solution.

(a) $\bar{x} = \dfrac{30}{5} = \boxed{6}$; deviations $-4, -2, 0, 2, 4$ give squares summing to 40, so $s^2 = \dfrac{40}{5-1} = \boxed{10}$ ($s \approx 3.16$).

(b) One *degree of freedom* is used up estimating the mean; dividing by $n-1$ removes the downward bias that dividing by n would introduce.

(c) "Unbiased" is a statement about the *long run*: averaged over many samples, $\bar{x}$ equals the true mean μ. Any single sample can still land above or below it.

Reminder

Distinguish the *estimator* (the rule) from the *estimate* (one number). Unbiasedness is a property of the rule across repeated sampling.

Exercise 8 $n = 25$, $\sigma = 15$, $\bar{x} = 100$.

Solution.

(a) $\text{SE} = \dfrac{15}{5} = 3$, so the interval is $100 \pm 1.96(3) = 100 \pm 5.88 = \boxed{[94.1,\ 105.9]}$.

(b) $n = 100$ gives $\text{SE} = 1.5$, so the width $\boxed{\text{halves}}$ (about ± 2.94).

(c) It describes the *procedure*: over repeated samples, 95% of such intervals contain the true mean. It is not a probability about this one interval, which either contains μ or does not.

(d) $\boxed{\text{Wider}}$. Catching the truth a larger fraction of the time (99%) needs a bigger net, so the multiplier grows ($1.96 \to 2.58$).

Reminder

A 95% CI for a mean (known σ) is $\bar{x} \pm 1.96\,\sigma/\sqrt{n}$: it narrows with more data and widens with more confidence.

Exercise 9 $H_0 : \mu = 50,\ \sigma = 10,\ n = 100,\ \bar{x} = 52.$

Solution.

(a) $\mathrm{SE} = \dfrac{10}{\sqrt{100}} = 1$, so $z = \dfrac{52-50}{1} = \boxed{2.0}$.

(b) $|z| = 2.0 > 1.96$, so $\boxed{\text{reject } H_0}$ at $\alpha = 0.05$ (two-sided $p \approx 0.046$).

(c) No. Significance also depends on n: with a large sample even a tiny, unimportant difference becomes significant, so report the effect size too.

(d) No. Failing to reject is *absence of evidence against* H_0, not proof that μ is exactly 50; the data are merely compatible with it.

Reminder

Test statistic $= \dfrac{\text{estimate} - \text{null value}}{\text{SE}}$. A test can reject H_0 or fail to; it never *proves* H_0.

Exercise 10 *Errors and power.*

Solution.

(a) *Type I:* reject H_0 when it is true (false alarm). *Type II:* fail to reject H_0 when H_1 is true (miss).

(b) The Type I error rate is $\boxed{5\%}$ (this is what α sets).

(c) Larger sample, larger true effect, higher α, or less noise.

(d) Tightening α to 0.01 lowers the false-alarm rate but *raises* the Type II rate and so *lowers* power: the two error types trade off, and you cannot shrink both at once without more data.

Reminder

α is the false-alarm rate you accept; power $= 1 - \beta$ is the chance of catching a real effect. Reducing one error type inflates the other unless n grows.

Exercise 11 *Prevalence* 5%*; sensitivity* 80%*; false-positive rate* 10%*.*

Solution.

(a) $\mathbb{P}(+) = 0.80(0.05) + 0.10(0.95) = 0.04 + 0.095 = \boxed{0.135}$.

(b) $\mathbb{P}(D \mid +) = \dfrac{0.04}{0.135} \approx \boxed{0.30}$.

(c) Both are true because they answer different questions. Sensitivity is $\mathbb{P}(+ \mid D)$; the posterior $\mathbb{P}(D \mid +)$ also weighs the rare base rate, so the 10% false positives from the large healthy group swamp the genuine ones.

(d) $\boxed{\text{Screening to a higher-risk group}}$. When disease is rare the base rate dominates, so raising the prior lifts $\mathbb{P}(D \mid +)$ more than a modest gain in test accuracy would.

> **Reminder**
>
> $\mathbb{P}(D \mid +) = \dfrac{\mathbb{P}(+ \mid D)\mathbb{P}(D)}{\mathbb{P}(+)}$: the prior (base rate) matters as much as the test's accuracy.

Exercise 12 *Bayes for events.*

Solution.

(a) $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(B \mid A)\,\mathbb{P}(A)}{\mathbb{P}(B)}$.

(b) $\dfrac{0.8(0.25)}{0.4} = \boxed{0.5}$.

(c) The belief $\boxed{\text{doubled}}$ $(0.5/0.25 = 2)$. That factor is $\dfrac{\mathbb{P}(B \mid A)}{\mathbb{P}(B)} = \dfrac{0.8}{0.4} = 2$: B is twice as likely under A as it is overall, so seeing B doubles A's probability.

(d) A higher base rate $\boxed{\text{raises}}$ $\mathbb{P}(\text{disease} \mid +)$.

> **Reminder**
>
> Bayes flips a known conditional $\mathbb{P}(B \mid A)$ into the one you want, $\mathbb{P}(A \mid B)$; the update factor is how much more likely B is under A than in general.

Exercise 13 *The modelling pipeline.*

Solution.

(a) A cognitive model proposes *how* the data were generated, a mechanism with parameters, rather than only describing them.

(b) Maximum likelihood carries out the $\boxed{\text{fit}}$ step (choosing θ).

(c) e.g. a $\boxed{\text{learning rate}}$, a decision threshold, a perceptual bias, or a memory-decay constant.

(d) The $\boxed{\text{standard error and confidence intervals}}$ (and, more generally, the posterior distribution) quantify the uncertainty in the fitted parameters.

Reminder

Pipeline: data → model $p(x \mid \theta)$ → fit → uncertainty → compare. Today's tools slot straight into it.

Exercise 14 *MLE versus MAP.*

Solution.

(a) MLE maximises the $\boxed{\text{likelihood}}$; MAP maximises the $\boxed{\text{posterior}}$ (likelihood × prior).

(b) They coincide when the prior is $\boxed{\text{flat}}$ (or when the data are so plentiful the likelihood dominates).

(c) With little data, $\boxed{\text{MAP}}$ is pulled more strongly by the prior.

(d) With scarce data a sensible prior *regularises* the estimate, for example preventing $\hat{p} = 0$ after 0 successes in 3 trials. It protects against overconfident conclusions from tiny samples.

Reminder

MAP is the MLE nudged by the prior. With a flat prior or abundant data, MAP → MLE; with little data, the prior does useful work.